



JKBOSE 2026

CLASS 11TH

PERMUTATION AND COMBINATIONS

MATHS

LECTURE-08

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TOPICS TO BE COVERED

1

Miscellaneous Exercise

2

Related Problems

A large, 3D rectangular frame with a red face and yellow borders, tilted slightly to the right. It has a dashed pink border around the text area.

**MISCELLANEOUS
EXERCISE ON
CHAPTER 6**

QUESTION

AUE DGHTR

How many words, with or without meaning, each of 2 vowels and 3 consonants can be formed from the letters of the word DAUGHTER?

(Sol) Here, the Total number of Selections

$$= {}^3C_2 \times {}^5C_3$$

Since each selection consists of 2 vowels & 3 consonants i.e. consists of 5 letters which can be arranged among themselves in

$${}_5P_5 = 5! \text{ ways.}$$

Therefore, the required no. of words

$$= ({}^3C_2 \times {}^5C_3) \times 5!$$

$$= \frac{3!}{2!1!} \times \frac{5!}{3!2!} \times 120$$

$$= \frac{3 \times \cancel{2!}}{\cancel{2!}} \times \frac{5 \times \cancel{4} \times \cancel{3!}}{\cancel{3!} \times \cancel{2!}} \times 120$$

$$= 3 \times 10 \times 120$$

$$= 3600. \text{ Hence the result}$$

QUESTION

How many words, with or without meaning, can be formed using all the letters of the word EQUATION at a time so that the vowels and consonants occur together?

(Sol) Required No. of words = $5! \times 3! \times 2!$

$$= 120 \times 6 \times 2$$

$$= 120 \times 12$$

$$= 1440. \text{ Hence the result.}$$

E U A I O Q T N

QUESTION

A committee of 7 has to be formed from 9 boys and 4 girls. In how many ways can this be done when the committee consists of:

- (i) Exactly 3 girls?
- (ii) Atleast 3 girls?
- (iii) Atmost 3 girls?

(Sol) (i) Required No. of ways

$$= {}^4C_3 \times {}^9C_4$$

(ii) Required No. of ways

$$= ({}^4C_3 \times {}^9C_4) + ({}^4C_4 \times {}^9C_3)$$

(iii) Required No. of ways

$$= ({}^4C_3 \times {}^9C_4) + ({}^4C_2 \times {}^9C_5) + ({}^4C_1 \times {}^9C_6) + ({}^4C_0 \times {}^9C_7)$$

Hence the result.

QUESTION



If the different permutations of all the letter of the word **EXAMINATION** are listed as in a dictionary, how many words are there in this list before the first word starting with E?

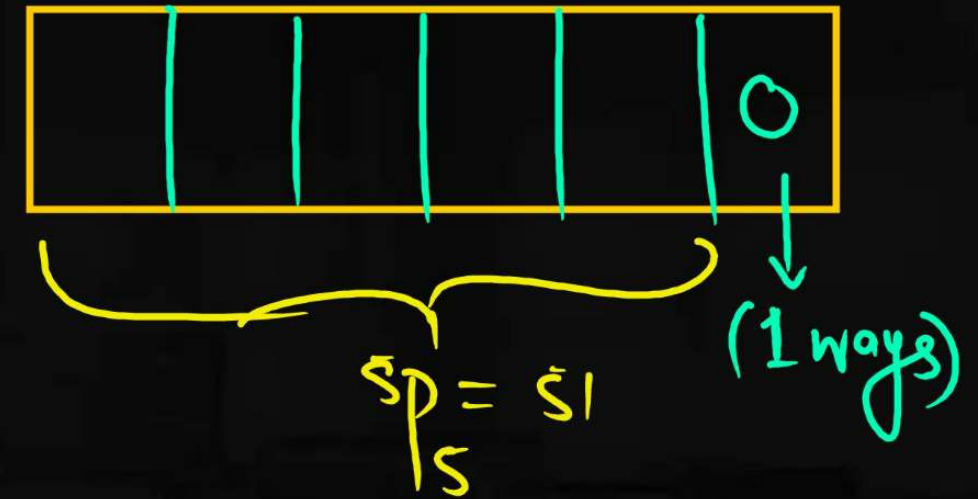
Homework.

QUESTION

How many 6-digit numbers can be formed from the digits 0, 1, 3, 5, 7 and 9 which are divisible by 10 and no digit is repeated?

$$5! \times 1$$

$$= 5! = 120.$$



QUESTION

The English alphabet has 5 vowels and 21 consonants. How many words with two different vowels and 2 different consonants can be formed from the alphabet?

(Sol) Required No. of words
$$= ({}^5C_2 \times {}^{21}C_2) \times 4!$$

Hence the result

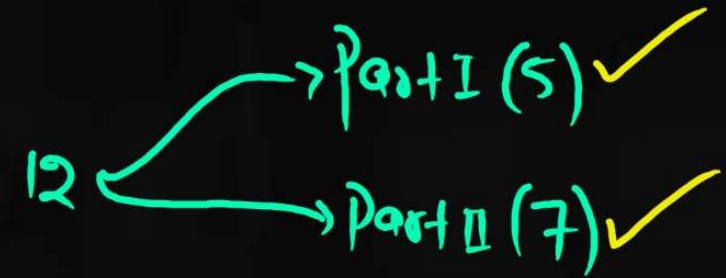
QUESTION

In an examination, a question paper consists of 12 questions divided into two parts i.e., Part I and Part II, containing 5 and 7 questions, respectively. A student is required to attempt 8 questions in all, selecting at least 3 from each part. In how many ways can a student select the question?

(Sol) Total no. of ways (selections)

$$= \binom{5}{3} \times \binom{7}{5} + \binom{5}{4} \times \binom{7}{4} + \binom{5}{5} \times \binom{7}{3}$$

Hence the result



QUESTION

Determine the number of 5-card combinations out of a deck of 52 cards if each selection of 5 cards has exactly one king.

$${}^4C_1 \times {}^{48}C_4$$

QUESTION

It is required to seat 5 men and 4 women in row so that the women occupy the even places. How many such arrangements are possible?

$$\left({}^4C_4 \times 4! \right) \times \left({}^5C_5 \times 5! \right)$$

$$= 24 \times 120$$

$$= ?$$

M_1	W_1	M_2	W_2	M_3	W_3	M_4	W_4	M_5
1	2	3	4	5	6	7	8	9

QUESTION

From a class of 25 students, 10 are to be chosen for an excursion party. There are 3 students who decide that either all of them will join or none of them will join. In how many ways can the excursion party be chosen?

(Sol) Required Number of selections

$$= \binom{3}{3} \times {}^{22}C_7 + \binom{3}{0} \times {}^{22}C_{10}$$
$$= {}^{22}C_7 + {}^{22}C_{10} \quad \text{Hence the result.}$$

QUESTION

In how many ways can the letters of the word **ASSASSINATION** be arranged so that all the S's are together?

SSSSAAIINATION

$$\frac{10!}{3!2!2!} \times \frac{4!}{4!} = \frac{10!}{3!2!2!}$$

Hence the result.

QUESTION

There are **10 points** in a row. In how many ways can **4 points** be selected such that **no two** of them are consecutive is



1 ✓✓ 35

$$\frac{6!}{6!} \times 7 \times \frac{4!}{4!} = {}^7C_4$$

2 ✗ 36

$$= {}^7C_4 = \frac{7!}{4!3!} = \frac{7 \times 6 \times 5 \times 4!}{4!3!}$$

3 ✗ 56

$$= 35$$

4 ✗ 63



No. of ways of selecting r points out of n points such that no two points are consecutive = $\frac{n-r+1}{C_r}$

QUESTION

$$\frac{4!}{2!2!} = \frac{4 \times 3 \times 2!}{2! \times 2!}$$

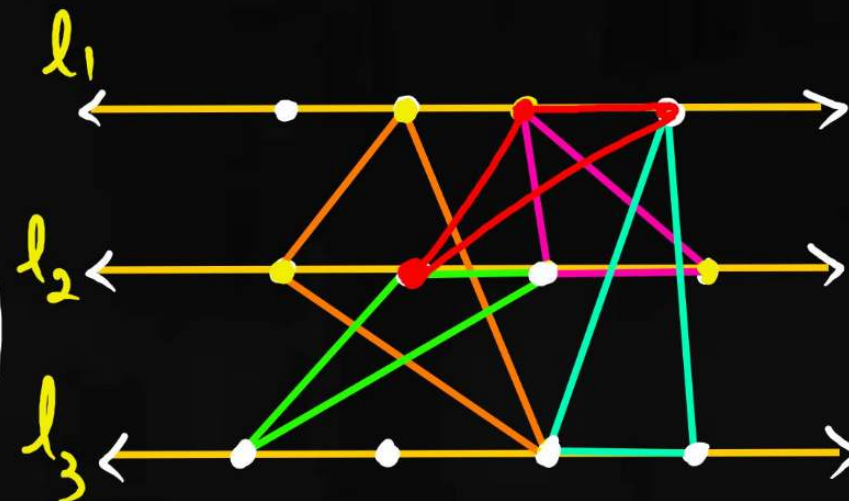


There are **three** Parallel lines. If any **4 points** are taken on **each of the lines**, then the total **number of triangles** with vertices at these points is :

- 1 145 ✗
- 2 144 ✗
- 3 208 ✓
- 4 None of these ✗

(Sol) Required No. of Triangles

$$\begin{aligned} &= ({}^4C_1 \times {}^4C_1 \times {}^4C_1) + ({}^3C_2 \times {}^4C_1 \times {}^4C_2 \times 2) \\ &= (4 \times 4 \times 4) + (3 \times 4 \times 6 \times 2) \\ &= 64 + 144 \\ &= 208 \end{aligned}$$



QUESTION

$${}^p C_2 = \frac{p(p-1)(\cancel{p-2})!}{(\cancel{p-2})! 2!}$$



There are **three** Parallel lines. If any **p points** are taken on **each of the lines**, then the total **number of triangles** with vertices at **these points** is :

☒ 1 $3p^2(p-1) + 1$ ✗

☒ 2 $3p^2(p-1)$ ✗

☒ 3 $p^2(4p-3)$ ✓

☒ 4 None of these

(Sol) Total no. of Triangles

$$= ({}^p C_1 \times {}^p C_1 \times {}^p C_1) + ({}^3 C_2 \times {}^p C_1 \times {}^p C_1 \times 2)$$

$$= (p \times p \times p) + \left(3 \times p \times \frac{p(p-1)}{2} \times 2 \right)$$

$$= p^3 + 3p^2(p-1)$$

$$= p^3 + 3p^3 - 3p^2$$

$$= 4p^3 - 3p^2 = p^2(4p-3)$$

QUESTION

The number of ways in which 5 different books can be distributed among 10 people if each person can get at most one book is :

1 $252 \times$

${}^{10}C_5 \times 5!$

2 $10^5 \times$

$B_1 B_2 B_3 B_4 B_5$

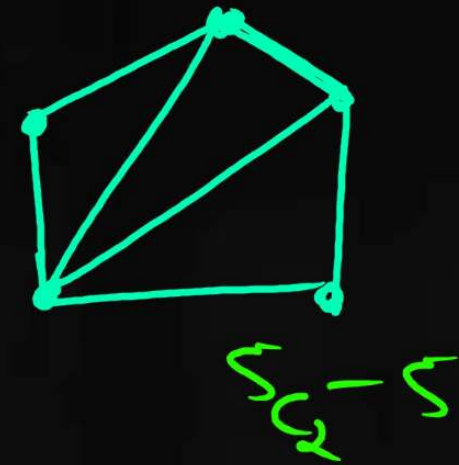
3 $5^{10} \times$

4 ${}^{10}C_5 \times 5!$ ✓✓

QUESTION(NCERT-EXAMPLER+NDA 2015)

If a Convex Polygon has 44 diagonals, then the number of sides of that Polygon is

- 1 11 ✓ $n_c - n = 44$
- 2 8 ✗ $\Rightarrow \frac{n(n-1)}{2} - n = 44$
- 3 15 ✗ $\Rightarrow n^2 - n - 2n = 88$
- 4 9 ✗ $\Rightarrow n^2 - 3n - 88 = 0$
 $\Rightarrow (n-11)(n+8) = 0$
 $\Rightarrow n = 11, -8$



QUESTION(NDA 2015)

✓ Prof. AMIN SOFI ✓ function.
✓ Prof. AZIZ UL AZEEM ✓ Complex Analysis



The number of ways in which 3 holiday tickets can be given to 20 employees of an organization, if each employee is eligible for any one or more of the tickets, is

☒ 1 1140

☒ 2 3420

☒ 3 6840

☒ 4 8000

$$20 \times 20 \times 20 \\ = 8000$$

$T_1 \rightarrow (20 \text{ ways})$
 $T_2 \rightarrow (20 \text{ ways})$
 $T_3 \rightarrow (20 \text{ ways})$

QUESTION(NDA 2024)



$${}^4C_2 = \frac{4!}{2!2!} = \frac{24}{4} = 6$$

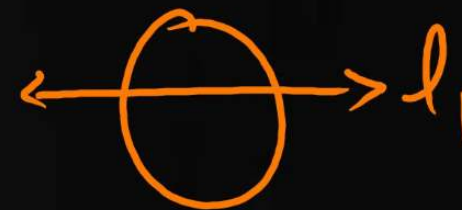
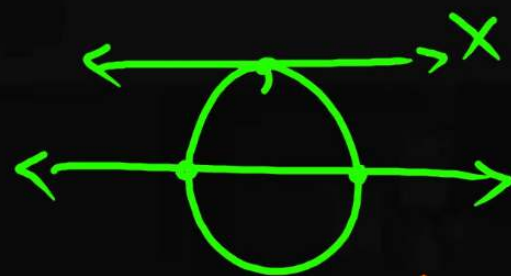
What is the maximum number of possible points of intersection of **four straight lines** and a **Circle** (intersection is between lines as well as circle and lines) ?

1 6 ✗

2 10 ✗

3 14 ✓

4 16 ✗



MSc in MATHEMATICS
+ PhD (Mathematical Biology)

$$\begin{aligned} & {}^4C_2 + {}^4C_1 \times 2 \\ &= 6 + 4 \times 2 \\ &= 6 + 8 = 14 \end{aligned}$$

$U! \times S!$
 $\text{MDHT} \rightarrow \text{DMHT.}$
 AEIOU DHT

$\eta_{pr} = \eta_{cs} \times \alpha!$
 $\boxed{\gamma = 1}$
Thank
You